WORKED SOLUTION

# STEP III 2000, Question 9

Mechanics • Collision

**STEP III 2000, Question 9**

Read the question before you go on. It is in the STEP Support Programme's question database:

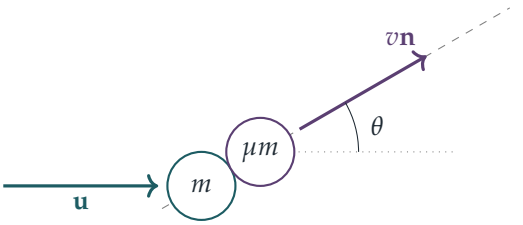
step.maths.org/questions/00-s3-q9

## The key idea

The discs are smooth, so during the impact the only force between them acts along the line of centres. The disc that starts at rest therefore moves off along that line, so $\mathbf{n}$ *is* the line of centres. Write its velocity as $v\mathbf{n}$, use conservation of momentum (a vector equation) to get the other disc's velocity, and then let conservation of kinetic energy fix $v$. Part (ii) is then one more energy equation, and the inequality comes from insisting that a cosine is at most 1.

## Solution

We model the discs as smooth particles on a smooth horizontal plane, so there is no friction between them or with the surface. Let $\theta$ be the angle between $\mathbf{u}$ and $\mathbf{n}$, and write $u = |\mathbf{u}|$, so that $\mathbf{u} \cdot \mathbf{n} = u\cos\theta$.




At the moment of impact the line of centres (dashed) points in the direction $\mathbf{n}$, making angle $\theta$ with the direction of $\mathbf{u}$ (dotted). The figure is only schematic; we do not need it for the algebra.

### Part (i)

The disc of mass $\mu m$ starts at rest and the only horizontal force on it during the impact is the contact force from the other disc, which acts along the line of centres. Since it moves off in direction $\mathbf{n}$, its velocity afterwards is $v\mathbf{n}$ for some speed $v > 0$.

Let the disc of mass $m$ have velocity $\mathbf{w}$ afterwards. Conservation of momentum gives

$$m\mathbf{u} = m\mathbf{w} + \mu m v\,\mathbf{n} \qquad \Longrightarrow \qquad \mathbf{w} = \mathbf{u} - \mu v\,\mathbf{n}.$$

The collision is perfectly elastic, so kinetic energy is conserved:

$$\tfrac{1}{2}m\,\mathbf{u}\cdot\mathbf{u} = \tfrac{1}{2}m\,\mathbf{w}\cdot\mathbf{w} + \tfrac{1}{2}\mu m v^2.$$

Using $\mathbf{n}\cdot\mathbf{n} = 1$,

$$\mathbf{w}\cdot\mathbf{w} = (\mathbf{u} - \mu v\mathbf{n})\cdot(\mathbf{u} - \mu v\mathbf{n}) = u^2 - 2\mu v\,\mathbf{u}\cdot\mathbf{n} + \mu^2 v^2.$$

Substituting and cancelling $\tfrac{1}{2}m$ and $u^2$,

$$0 = -2\mu v\,\mathbf{u}\cdot\mathbf{n} + \mu^2 v^2 + \mu v^2 = \mu v((1+\mu)v - 2\,\mathbf{u}\cdot\mathbf{n}).$$

Now $\mu > 0$, and $v \neq 0$ because the struck disc does move. So we may divide by $\mu v$ to get

$$v = \frac{2\,\mathbf{u}\cdot\mathbf{n}}{1+\mu},$$

which is the required speed. As a by-product, $v > 0$ tells us that $\mathbf{u}\cdot\mathbf{n} > 0$, so $0 \leq \theta < \tfrac{1}{2}\pi$.

**Ask yourself** Why don't we need Newton's experimental law here?

We could use it: with $e = 1$, the components of relative velocity along $\mathbf{n}$ are reversed, and the components perpendicular to $\mathbf{n}$ are unchanged because the discs are smooth. But "perfectly elastic" means kinetic energy is conserved, and that single scalar equation, with momentum, already pins down $v$. The working above is exactly equivalent and avoids splitting into components.

#### Part (ii)

The total kinetic energy after the collision equals the initial $\tfrac{1}{2}mu^2$. If the two discs share it equally, each has $\tfrac{1}{4}mu^2$, so for the struck disc

$$\tfrac{1}{2}\mu m v^2 = \tfrac{1}{4}mu^2 \qquad \Longrightarrow \qquad \mu v^2 = \tfrac{1}{2}u^2.$$

Using part (i) with $\mathbf{u}\cdot\mathbf{n} = u\cos\theta$, we have $v = \dfrac{2u\cos\theta}{1+\mu}$, so

$$\frac{4\mu u^2\cos^2\theta}{(1+\mu)^2} = \frac{u^2}{2} \qquad \Longrightarrow \qquad \cos^2\theta = \frac{(1+\mu)^2}{8\mu}.$$

Since $\cos\theta > 0$ (from the end of part (i)), we take the positive root:

$$\cos\theta = \frac{1+\mu}{2\sqrt{2\mu}}.$$

For such an angle to exist we need $\cos\theta \le 1$, that is $\cos^2\theta \le 1$:

$$\frac{(1+\mu)^2}{8\mu} \le 1 \iff (1+\mu)^2 \le 8\mu \qquad \text{(multiplying by } 8\mu > 0\text{)}$$
$$\iff \mu^2 - 6\mu + 1 \le 0.$$

The quadratic $\mu^2 - 6\mu + 1$ has roots $\mu = 3 \pm \sqrt{9-1} = 3 \pm \sqrt{8}$, and it is negative or zero exactly between them. Hence

$$3 - \sqrt{8} \le \mu \le 3 + \sqrt{8},$$

as required. (The lower bound is positive, $3 - \sqrt{8} \approx 0.172$, so it is consistent with $\mu > 0$.) Conversely, for every $\mu$ in this range the formula gives a valid cosine in $(0, 1]$, so equal sharing really is possible; at either end $\theta = 0$, a head-on collision.

## Checks

- Head-on case: if $\mathbf{n}$ is parallel to $\mathbf{u}$, part (i) gives $v = 2u/(1+\mu)$, the familiar one-dimensional elastic result. With $\mu = 1$ this is $v = u$: the moving disc stops and passes all its motion on, as in snooker.
- Equal masses, $\mu = 1$: part (ii) gives $\cos\theta = 2/(2\sqrt{2}) = 1/\sqrt{2}$, so $\theta = 45°$. This fits the classic fact that equal smooth discs move off at right angles after an elastic collision; equal energies then force the two speeds to be equal, so each makes $45°$ with $\mathbf{u}$.
- Numerically, we simulated 10 000 random smooth elastic collisions (impulse along $\mathbf{n}$), confirmed conservation of momentum and energy and the speed in part (i), and confirmed equal kinetic energies at the angle found in part (ii) for $\mu$ between 0.172 and 5.83. For example, $\mu = 0.1$ gives $\cos^2\theta = 1.51 > 1$, which is impossible.

## Common slips

- Treating $\mathbf{u}$ and $\mathbf{n}$ as parallel. The collision is oblique; only the component $\mathbf{u} \cdot \mathbf{n}$ along the line of centres is transferred.
- Dividing by $v$ without saying why it is non-zero ($v = 0$ is the "no collision" solution of the energy equation).
- Forgetting to rule out the negative square root for $\cos\theta$.

## Beyond STEP

The trick of writing the struck particle's velocity along the impulse and then using momentum and energy is the heart of elastic scattering in physics. It is cleanest in the centre-of-mass frame, where an elastic collision just rotates both velocities without changing their sizes; the angle range in part (ii) then becomes a statement about which

lab-frame angles are reachable. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions in both frames very clearly.

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